

ResearchLens - Autonomous Academic Synthesis

Research Question: What are the core scaling laws and architectural innovations enabling modern Trans...

Date: 2026-10-08 | Confidence: MEDIUM | Citation Integrity: 0.0% | Papers Analyzed: 3

Executive Summary

Attention Is All You Need was introduced by Ashish Vaswani, Noam Shazeer, Niki Parmar et al. in 2017.

Methodological Comparison

Paper	Year	Venue	Citations
Attention Is All You Need	2017	NeurIPS	95,000
Scaling Laws for Neural Language Models	2020	arXiv	4,200
FlashAttention: Fast and Memory-Efficient Exact Attention w...	2022	NeurIPS	2,100

Detailed Research Findings

Core Scaling

Language model test performance follows predictable power-law scaling relationships [2].

Additional Empirical Findings

Transformers eliminate sequential recurrence in favor of multi-head self-attention mechanisms [1]. IO-aware tiling in FlashAttention provides 2-4x speedups without approximation [3].

Limitations & Boundary Conditions

- Retrieval & Evidence Funnel: 120 discovered -> 85 unique -> 18 relevant -> 12 full text -> 340 passages -> 3 evidence-bearing -> 3 verified claims.
- Citation integrity (0.0%) is below the 80% threshold; 4 unsupported sentence(s) were pruned by NLI audit.

Conclusion

In summary, empirical literature analysis addresses: "What are the core scaling laws and architectural innovations enabling modern Transformer models?". Foundational anchor paper: 'Attention Is All You Need' (2017). Anchor confidence: High. Findings verified with 0.0% citation integrity. Warnings: 4 claim sentence(s) removed due to lack of verifiable evidence..

References

- [1] Attention Is All You Need (2017). Ashish Vaswani, Noam Shazeer, Niki Parmar et al. NeurIPS. DOI: 10.48550/arXiv.1706.03762
- [2] Scaling Laws for Neural Language Models (2020). Jared Kaplan, Sam McCandlish, Tom Henighan et al. arXiv. DOI: 10.48550/arXiv.2001.08361
- [3] FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness (2022). Tri Dao, Daniel Y. Fu, Stefano Ermon et al. NeurIPS. DOI: 10.48550/arXiv.2205.14135